BHUBESHNATH P

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EXECUTIVE SUMMARY

Biotechnology undergraduate with hands-on experience integrating AI/ML with laboratory research, data analysis, and software tools. Proficient in Python, MATLAB, and research-grade analytical instruments. Strong technical communication skills developed through academic projects and collaborative research. Quick learner with demonstrated ability to troubleshoot complex scientific and computational problems directly transferable to technical support roles.

EDUCATIONAL CREDENTIALS

Bachelor of Technology in Biotechnology, *Rajalakshmi Engineering College* — 2022 – 2026 | Chennai

CGPA : 8.02 | GATE (BT) 2025 Qualified

Relevant coursework: Industrial Biotechnology, Nanotechnology in Agriculture, Food science and Technology.

SKILLS

- **Programming & Data:** Python, MATLAB, SQL, R, NumPy, Pandas, Matplotlib, Data Analysis.
- **Microscopy & Microbiology:** Crystal Violet Biofilm Assay, Live/Dead Staining, DAPI Staining, Antimicrobial Assays.
- **Analytical Instrumentation:** UHPLC , NanoDrop UV-Vis Spectrophotometry, RT-PCR, Fluorescent microscope.
- **Nanotechnology & Formulation:** Green Nanoparticle Synthesis & Characterization, Phytochemistry, Product Formulation.
- **Bioinformatics & Computational Biology:** Molecular Docking (AutoDock, PyRx), Transcriptomic Analysis (Scanpy, GEO datasets).

INTERNSHIPS AND TRAINING

Research Intern, *DST-FIST Sponsored Analytical Instruments Lab, Rajalakshmi Engineering College* — 06/2024 – 07/2024

- Detected SARS-CoV-2 via multiplex RT-PCR with Ct value interpretation.
- Ran fluorescence microscopy assays: biofilm formation (Crystal Violet), live/dead bacterial viability (SYTO9/PI), and DAPI nuclear staining.
- Quantified caffeine in beverages via UHPLC (C18 column, calibration-curve analysis).
- Synthesized and characterized green silver nanoparticles via NanoDrop UV-Vis spectrophotometry.

Artificial Intelligence, *Corizo* — 03/2024 – 04/2024

- Explored AI model development with applications in biotechnology.
- Hands-on exposure to supervised learning, model evaluation, and data pipelines.

AIML approaches for drug delivery for cancer, *Institute of Innovation* — 01/2024 – 01/2024

- Applied machine learning techniques to optimize cancer drug delivery systems.
- Understood how computational tools interface with real-world biomedical problems.

PROJECTS

Wound Chron, *Model-Based Estimation of Wound Aging Using Plant-Derived Nanoparticles and Bioinformatics Tools*

- Green-synthesized ZnO nanoparticles (from Ageratum conyzoides extract), characterized via UV-Vis, XRD, and FTIR spectroscopy.
- Engineered a JavaScript-based image-analysis pipeline (Gaussian blur, Otsu thresholding, healing-rate estimation).
- Applied molecular docking (quercetin, −8.3 kcal/mol vs. target 3MME) and GEO transcriptomic biomarker analysis (Scanpy) to computationally support the wound-aging model.
- Translated findings into a deployed digital wound-monitoring application (bhubeshbtech.github.io/WoundChron/) .

Nyctan Chews, *Enhanced Bioavailability of Nyctanthes arbor tristis Bioactives through gummy formulation*

- Applied a structured design-thinking process to formulate and validate a plant-based nutraceutical product, backed by cost and sustainability (SDG) analysis.
- Validated physicochemical properties (pH ~4.2, °Brix ~36°, moisture 18–20%) and conducted preliminary antimicrobial testing.

COURSES AND CERTIFICATIONS

- Python, HCL GUVI
- SQL for Healthcare Professionals
- Artificial Intelligence
- Python Data Analysis for Healthcare
- Excel for Healthcare: Practical Applications
- Machine Learning Fundamentals for Healthcare